

strategiesDataAware.pdf



abcd__1234



Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



[Accede al documento original](#)

 **Gemini 3**
Nuestro modelo más preciso

Sintetiza horas de
Investigación en minutos

Necesito estudiar a fondo el comportamiento de la fotosíntesis según el tipo de planta y el entorno



 Deep Research 



 Fuentes 

 Archivos

Estudia con Gemini

Expert

2 DÍAS
CONSECUTIVOS**86€**

Long expert

5 DÍAS
DE LIBRE USO**185€**Más info.:
elsubidon.es

Estat	Acabat
Començat el	dijous, 4 de desembre 2025, 20:50
Completat el	dijous, 4 de desembre 2025, 20:58
Durada	7 minuts 46 segons
Qualificació	7,00 sobre 7,00 (100%)

Pregunta 1

Completa

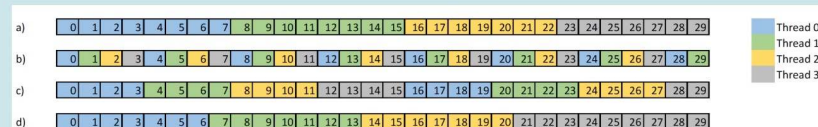
Puntuació 1,00
sobre 1,00 Marca la
pregunta

Given the following parallel region in OpenMP initializing vector a:

```
#define n 30
int vector[n];

#pragma omp parallel num_threads(4)
{
    int myid = omp_get_thread_num( );
    int howmany = omp_get_num_threads( );
    int num_elems = n / howmany;
    int start = myid * num_elems;
    int end = start + num_elems;
    if ( myid == howmany-1 ) end = n;
    for ( i = start; i < end; i++ ) {
        vector[i] = foo(i);
    }
}
```

Indicate which one of the following OUTPUT data decompositions for vector is being applied to drive the parallelization of the code using implicit tasks:



Note: a) corresponds to a block data decomposition in which threads 0 and 1 have 8 consecutive elements and threads 2 and 3 only 7; b) corresponds to a cyclic data decomposition, with the first element assigned to thread 0; c) corresponds to a block-cyclic data decomposition with block size equal to 4; and finally, d) corresponds to a block data decomposition in which threads 0, 1 and 2 have 7 consecutive elements each and thread 3 has 9.

Note: a) corresponds to a block data decomposition in which threads 0 and 1 have 8 consecutive elements and threads 2 and 3 only 7; b) corresponds to a cyclic data decomposition, with the first element assigned to thread 0; c) corresponds to a block-cyclic data decomposition with block size equal to 4; and finally, d) corresponds to a block data decomposition in which threads 0, 1 and 2 have 7 consecutive elements each and thread 3 has 9.

Tríe-u-ne una:

- ☐ Data decomposition a)
- ☐ Data decomposition b)
- ☐ Data decomposition c)
- ☒ Data decomposition d)

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00

 Marca la
pregunta

If in the previous code we change the line:

```
if ( myid == howmany-1 ) end = n;
```

to the following code sequence:

```
int rem_elems = n % howmany;  
if (rem_elems != 0) {  
    if (myid < rem_elems) {  
        start += myid;  
        end += (myid+1);  
    } else {  
        start += rem_elems;  
        end += rem_elems;  
    }  
}
```

indicate which one of the OUTPUT data decompositions shown in the first question would be applied to drive the parallelization of the code using implicit tasks:

Trieu-ne una:

- ☒ Data decomposition a)
- ☐ Data decomposition b)
- ☐ Data decomposition c)
- ☐ Data decomposition d)

SN 25/26

PARA LA COMUNIDAD
UNIVERSITARIA

EL SUBIDÓN

DE SIERRA NEVADA

Expert

2 DÍAS
CONSECUTIVOS

86€

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00

Marca la
pregunta

Long expert

5 DÍAS
DE LIBRE USO

185€

Más info.:
elsubidon.es



If the code is changed to:

```
#pragma omp parallel
{
    int myid = omp_get_thread_num( );
    int howmany = omp_get_num_threads( );
    for (int i = myid; i < n; i+=howmany) {
        vector[i] = foo(i);
    }
}
```

Indicate which one of the OUTPUT data decompositions shown in the first question is being applied to drive the parallelization of the code using implicit tasks:

Trieu-ne una:

- ☐ Data decomposition a)
- ☒ Data decomposition b)
- ☐ Data decomposition c)
- ☐ Data decomposition d)

WUOLAH

Pregunta 4

Completa

Puntuació 1,00
sobre 1,00

 Marca la
pregunta

How would you change the loop in the question 3 in order to follow the data decomposition that has not been applied in any of the previous three codes? You can assume that constant BS is defined as 4.

Trieu-ne una:

- ☐ I just simply need to change the i loop to go from myid*BS to n in steps of 1, as follows:

```
for (int i = myid*BS; i < n; i++)  
{ ... }
```

- ☐ I just simply need to change the i loop to go from myid*BS to n in steps of BS, as follows:

```
for (int i = myid*BS; i < n; i += BS)  
{ ... }
```

- ☐ I just simply need to change the i loop to go from myid*BS to n in steps of howmany*BS, as follows:

```
for (int i = myid*BS; i < n; i += howmany*BS)  
{ ... }
```

- ☒ I need to add a new loop to the previous code traversing the consecutive elements inside each block, as follows:

```
for (int ii = myid*BS; ii < n; ii += howmany*BS)  
  for (int i = ii; i < min(ii+BS, n); ii++)  
  { ... }
```


FIEP

FERIA
INTERNACIONAL
DE ESTUDIOS
DE POSTGRADO

www.fiep.es



Consigue **tu** **Beca**

Feria de Postgrado – FIEP Barcelona

Hotel NH Collection Constanza

jueves, 26 de febrero de 2026, 16:00 - 19:00



Entrada Gratuita. ¡Ven y participa en el Sorteo de
3 Becas de 10.000 € para tu postgrado!

INSCRÍBETE GRATIS

Pregunta 5

Completa

Puntuació 1,00
sobre 1,00 Marca la pregunta

Of the four proposed data decompositions for vector shown in the first question, and assuming that the execution time for function foo is constant, only data decompositions labeled a) and b) may introduce a load unbalance (measured in terms of the difference in the number of elements assigned to implicit tasks) of 1 data element. Data decomposition c) may have a maximum load unbalance (i.e. the largest difference between number of elements assigned to two different implicit tasks) equal to the size of the block while data decomposition d) a maximum load unbalance equal to the number of processors used minus 1.

Trieu-ne una:

- ☒ Verdader
- ☐ Fals

Pregunta 6

Completa

Puntuació 1,00
sobre 1,00 Marca la pregunta

If the parallel region is changed as follows in order to do a certain computation with the elements of vector:

```
#pragma omp parallel reduction(+:sum)
{
    int myid = omp_get_thread_num( );
    int howmany = omp_get_num_threads( );
    for (int i = 0; i < n; i++) {
        int j = swap(i, n);
        if (j%howmany == myid) sum += foo(vector[j], i);
    }
}
```

being swap a function that returns an integer in the range 0 to n-1 based on the value of i. Indicate which one of the INPUT data decompositions for vector shown in the first question is being applied to drive the parallelization of the code using implicit tasks:

Trieu-ne una:

- ☒ INPUT data decomposition b)
- ☐ OUTPUT data decomposition b)
- ☐ INPUT data decomposition c)
- ☐ OUTPUT data decomposition c)

Si estás en tu spending era...

mejor tener una app que te diga en qué
tiendas se ha quedado registrada tu tarjeta.
¡Como la app de ING!

Saber más



Pregunta 7

Completa

Puntuació 1,00
sobre 1,00

Marca la
pregunta

Finally, if the previous code is changed as follows:

```
#pragma omp parallel reduction(+:sum)
{
    int myid = omp_get_thread_num( );
    int howmany = omp_get_num_threads( );
    int num_elems = n / howmany;
    int start = myid * num_elems;
    int end = start + num_elems;
    if (myid == howmany-1) end = n;
    for (int i = 0; i < n; i++) {
        int j = swap(i, n);
        if ((j>=start) && (j<end)) sum += foo(a[j], i);
    }
}
```

Indicate which one of the INPUT data decompositions for vector a shown in the first question is being applied to drive the parallelization of the code using implicit tasks:

Trieu-ne una:

- ☐ INPUT data decomposition a)
- ☐ OUTPUT data decomposition a)
- ☒ INPUT data decomposition d)
- ☐ OUTPUT data decomposition d)



do your thing

WUOLAH